

Reasoning Termination, Not Reasoning Capability

Budget forcing on VibeThinker-3B, AIME 2025

Abu Ahmad · 2121725042 · CSE 465 · North South University

The finding

VibeThinker-3B scores **91.4** on AIME 2025 in its paper; we reproduced **80.8**. The gap is not weaker reasoning — the model solves the problem and then never writes the answer down. Of 120 sampled attempts, essentially every one that finished was correct, while every one cut off by the token budget scored zero for want of a boxed answer. Across 1,200 sample-budget observations the model volunteered **exactly one wrong answer**: it states the correct answer or states nothing at all. Interrupting it at the budget and forcing it to commit recovers **7 to 13 points**, has **never** destroyed a correct answer, and costs about five tokens.

Table 1 · Accuracy against token budget, without forcing

Budget	1k	2k	3k	4k	6k	8k	10k	12k	14k	16k	32k
Accuracy %	3.3	13.3	23.3	30.0	38.3	46.7	51.7	56.7	58.3	63.3	80.8
Cut off %	100	98.3	92.5	87.5	77.5	67.5	63.3	59.2	53.3	48.3	20.8

Accuracy and truncation are mirror images. This is the whole argument: what changes across the row is the budget, not the model.

Table 2 · Effect of budget forcing

Budget	No forcing	+ Forcing	Gain	McNemar p
4k	27.5%	40.8%	+13.3	—
8k	46.7%	55.8%	+9.1	0.0026
16k	63.3%	70.8%	+7.5	0.0077
32k	80.8%	not run	—	—

Table 3 · Does forcing ever destroy a correct answer?

Budget	Truncated	Rescued	Damaged	McNemar p
8k	81	11	0	0.0026
16k	58	9	0	0.0077

Zero damage across 139 interventions. Forcing only ever acts on a sample that had already failed; one that stopped on its own is left untouched.

Table 4 · What the 120 samples are made of

Condition	Correct	Wrong	No answer
8k, no forcing	56	0	64
8k, forced	67	53	0
16k, no forcing	76	0	44
16k, forced	85	33	2

The zero column is the sharpest result in the project: left alone, the model never guesses. Forcing converts silence into a commitment — and a commitment can be wrong.

Table 5 · Recovery rate among samples forcing can actually help

Budget	Truncated	Already correct	Rescuable	Rescued	Rate
8k	81	17	64	11	17.2%
16k	58	14	44	9	20.5%

Roughly one failed sample in five is recovered. Random guessing on AIME, where answers are integers from 0 to 999, would succeed 0.1% of the time — so forcing is about 150 times better than chance, and is reading something real out of the partial trace.

All values measured on a Kaggle Tesla T4, float16, vLLM — 30 problems, K = 4, 22.6 GPU-hours. Nothing is estimated; quantities that were not measured are marked as such. Code and data: github.com/abuahmad369/llm-budget-forcing